Governance-Aware Earth Observation Learning

Contract-Governed Training with Observed Service Agreement Graphs

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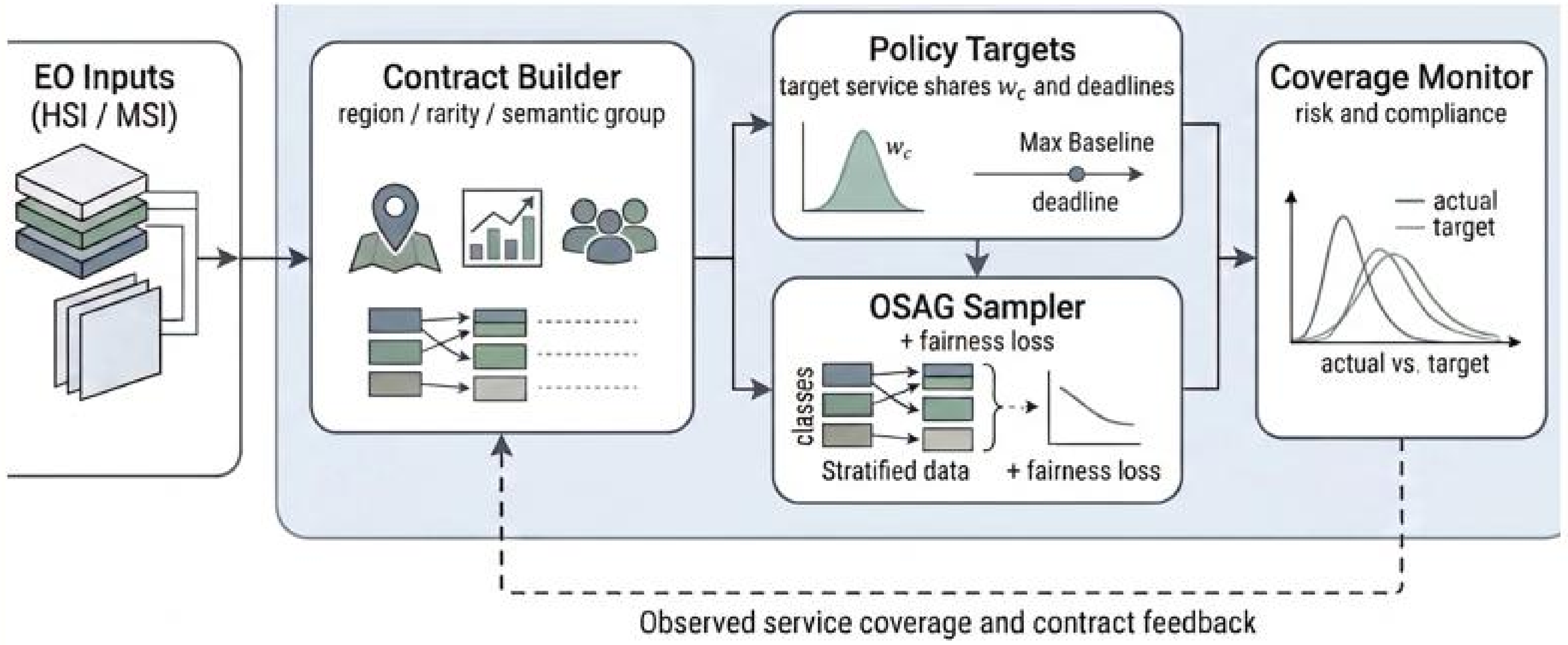
Thesis Defense Presentation



Final Figure 1.1 supporting visual

Service-aware EO training through explicit contract targets and observed coverage control.

Talk Roadmap

This presentation moves from the problem and framework to the benchmark evidence, then to the demo, code orientation, and claim boundaries.

1. Problem and Motivation

Why accuracy-only EO training misses the service-allocation question.

2. OSAG Framework and Benchmarks

Contracts, target shares, and the HSI/MSI evaluation design.

3. Results, Trade-offs, and Robustness

Fresh reruns, ablation, runtime interpretation, and a scoped backbone extension.

4. Demo, Code, and Boundaries

Dispatch behavior, repository orientation, limitations, and reproducibility.

Background and Motivation

Why the standard training view is incomplete

- Standard EO training usually optimizes aggregate predictive utility such as overall accuracy.
- Real EO services care about who receives attention during training: rare crops, vulnerable regions, or policy-critical semantic groups.
- If service allocation stays implicit, a model can look strong globally while underserving the contracts that matter most.
- This thesis treats training exposure as a governance object rather than as a side effect of data frequency.

Problem in one sentence

The key question is not only 'Does the model perform well on average?' but also 'Who is being served during training?'

Why governance is needed

Accuracy-only view

good global score

Can still ignore policy-critical strata if they are statistically small.

Governance view

explicit service

Tracks whether observed exposure matches the intended contract policy.

This thesis adds a lightweight governance layer instead of replacing the classifier backbone.

Research Gap and Guiding Questions

Central gap

EO training pipelines rarely expose service allocation explicitly. They optimize losses over samples, not compliance with a policy over contracts.

RQ1

How can EO training be reformulated so that service allocation over meaningful contracts becomes explicit and measurable?

RQ2

Can a lightweight governance layer drive empirical coverage toward target shares without requiring a new backbone?

RQ3

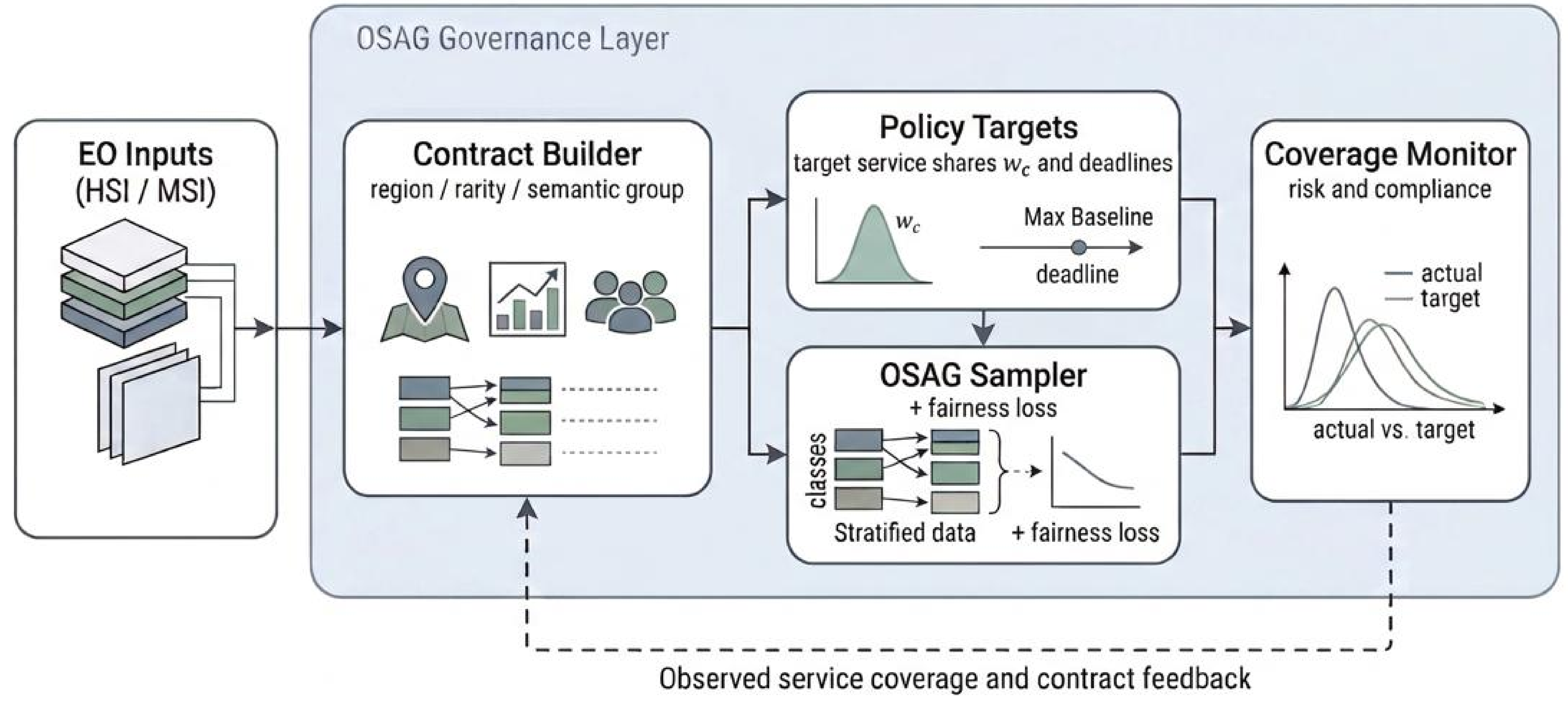
What is the empirical relationship between governance quality and predictive utility on HSI and MSI benchmarks?

RQ4

Can the thesis provide a runnable demo that communicates the governance mechanism clearly in a defense setting?

The deck follows the same arc: problem, method, fresh evidence, operational demo, and clear claim boundaries.

Figure 1.1: Method Overview



How to read the pipeline

- EO inputs provide sample metadata, labels, and contract-relevant attributes.
- The Contract Builder groups samples into service units such as grid cells, semantic groups, or rarity-aware groups.
- Policy Targets convert priorities and contract sizes into explicit target service shares.
- The OSAG Sampler injects the policy into training exposure, while the Coverage Monitor measures compliance.
- The feedback loop makes service allocation visible, auditable, and controllable during ordinary training.

Key point: OSAG wraps a standard EO classifier. It does not replace the backbone.

Core Concepts

Contract

A meaningful service unit, not just a class label. Example: dataset + grid cell + rarity flag.

Target service share

The intended fraction of training attention assigned to each contract.

Empirical coverage

The observed fraction of training steps actually spent on each contract.

PrioCovErr

The L1 distance between empirical coverage and target shares. Smaller means better policy compliance.

alpha

Mixing knob between the baseline sampler and pure OSAG. Higher alpha means stronger governance control.

lambda_C

Fairness-loss weight that adds pressure when high-priority contracts remain harder in loss space.

Benchmark and Contract Design

Formal benchmark inputs

- HSI main benchmark: corrected Indian Pines + corrected Salinas after common-band alignment.
- MSI main benchmark: canonical 13-band EuroSAT MSI from the formal local input archive.
- Main benchmark tables are fresh five-seed reruns from the formal script pipeline.
- EuroSAT contract-design ablation is a fresh three-seed rerun, scoped as an extension rather than a second main benchmark.

Realized contract schemas

Case	Schema	Count
HSI	dataset + 4x4 grid cell + rare flag	44
EuroSAT coarse	four semantic service groups	4
EuroSAT fine	semantic group + rare flag	6

Target-share policy

Target service share is proportional to priority times contract size.

- HSI: rare contracts use priority 3; non-rare contracts use priority 1.
- EuroSAT coarse: urban and water use priority 2; agricultural and natural use priority 1.
- EuroSAT fine: rare contracts use priority 2; non-rare contracts use priority 1.

Main HSI Benchmark Results

Indian+Salinas fresh five-seed summary

Policy	Acc_all	Acc_high	PrioCovErr
Random	91.47	82.97	23.49
Class-Balanced	90.25	86.26	31.37
OSAG-Mix (a=0.50)	91.68	86.51	11.75
OSAG	90.85	88.83	0.26
OSAG-FairLoss (l=0.5)	91.10	91.15	0.26

Values are percentages except PrioCovErr, which is shown in percentage points to match the thesis tables.

Governance gain

23.49 -> 0.26

OSAG-family operating points collapse HSI policy misalignment toward zero.

High-priority utility

82.97 -> 91.15

OSAG-FairLoss gives the strongest Acc_high while preserving near-zero PrioCovErr.

Why this matters

not just class balancing

Class-Balanced helps some utility metrics but still leaves PrioCovErr high at 31.37.

Main EuroSAT MSI Results

EuroSAT MSI fresh five-seed summary

Policy	Acc_all	Acc_high	PrioCovErr
Random	86.93	73.68	23.81
Class-Balanced	85.91	74.25	17.18
OSAG-Mix (a=0.75)	86.70	74.75	5.96
OSAG	86.39	76.05	0.20
OSAG-FairLoss (l=0.5)	86.30	77.04	0.20

Values are percentages except PrioCovErr, which is shown in percentage points to match the thesis tables.

Governance gain

23.81 -> 0.20

On MSI as well, OSAG moves policy error from about twenty-four points to almost zero.

High-priority utility

73.68 -> 77.04

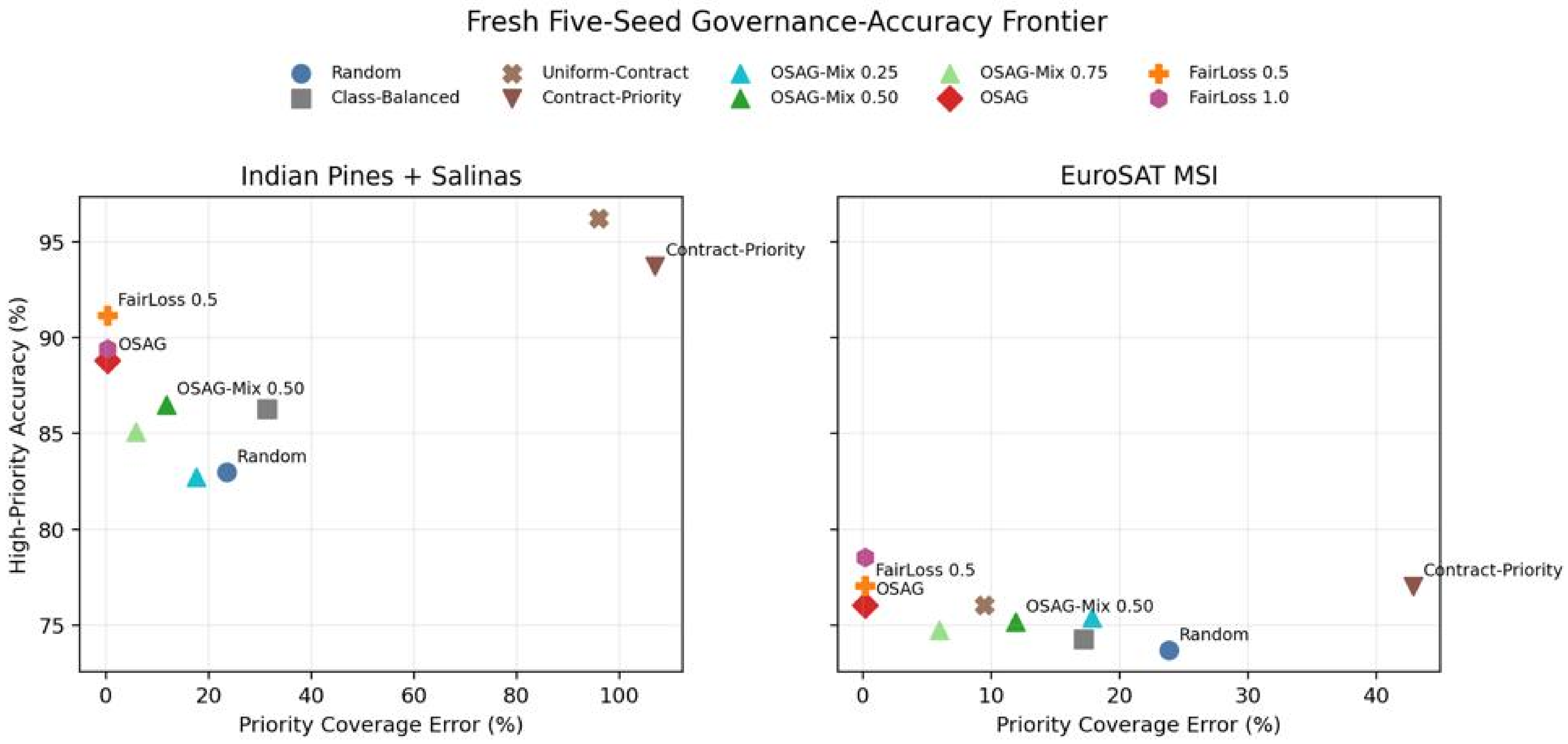
OSAG-FairLoss gives the best Acc_high among the highlighted operating points.

Cross-modality claim

HSI and MSI agree

The governance effect is not confined to one sensing modality or one contract schema.

Why Accuracy Alone Is Not Enough



Read Table 5.2 correctly

The reported deltas are absolute percentage-point differences relative to Random, not relative percentage changes.

Two direct examples

- HSI OSAG vs Random: Acc_all -0.62, Acc_high +5.85, PrioCovErr -23.23 points.
- EuroSAT OSAG vs Random: Acc_all -0.54, Acc_high +2.36, PrioCovErr -23.61 points.

Main message

Average accuracy remains important, but it cannot tell us whether the intended contracts were actually served during training.

Contract Design, Runtime, and Robustness

Contract-design ablation

Fresh three-seed EuroSAT rerun

Variant	Rnd Acc_all	OSAG Acc_all	Rnd Err	OSAG Err
Coarse	86.98	86.03	33.29	0.12
Fine	86.98	86.28	23.80	0.21

- Fine contracts start from a lower Random policy-error baseline.
- The accuracy cost of governance is smaller in the fine design.
- Contract design therefore changes governance cost.

Runtime interpretation

EuroSAT per epoch

1.367s vs 1.358s

Random and OSAG stay in the same practical runtime range.

HSI per epoch

3.524s vs 2.938s

The slower Random mean is driven mainly by one outlier seed, not by an OSAG speed advantage.

Thesis wording to keep: runtime should be interpreted as comparable-overhead evidence, not as a claim that OSAG is intrinsically faster.

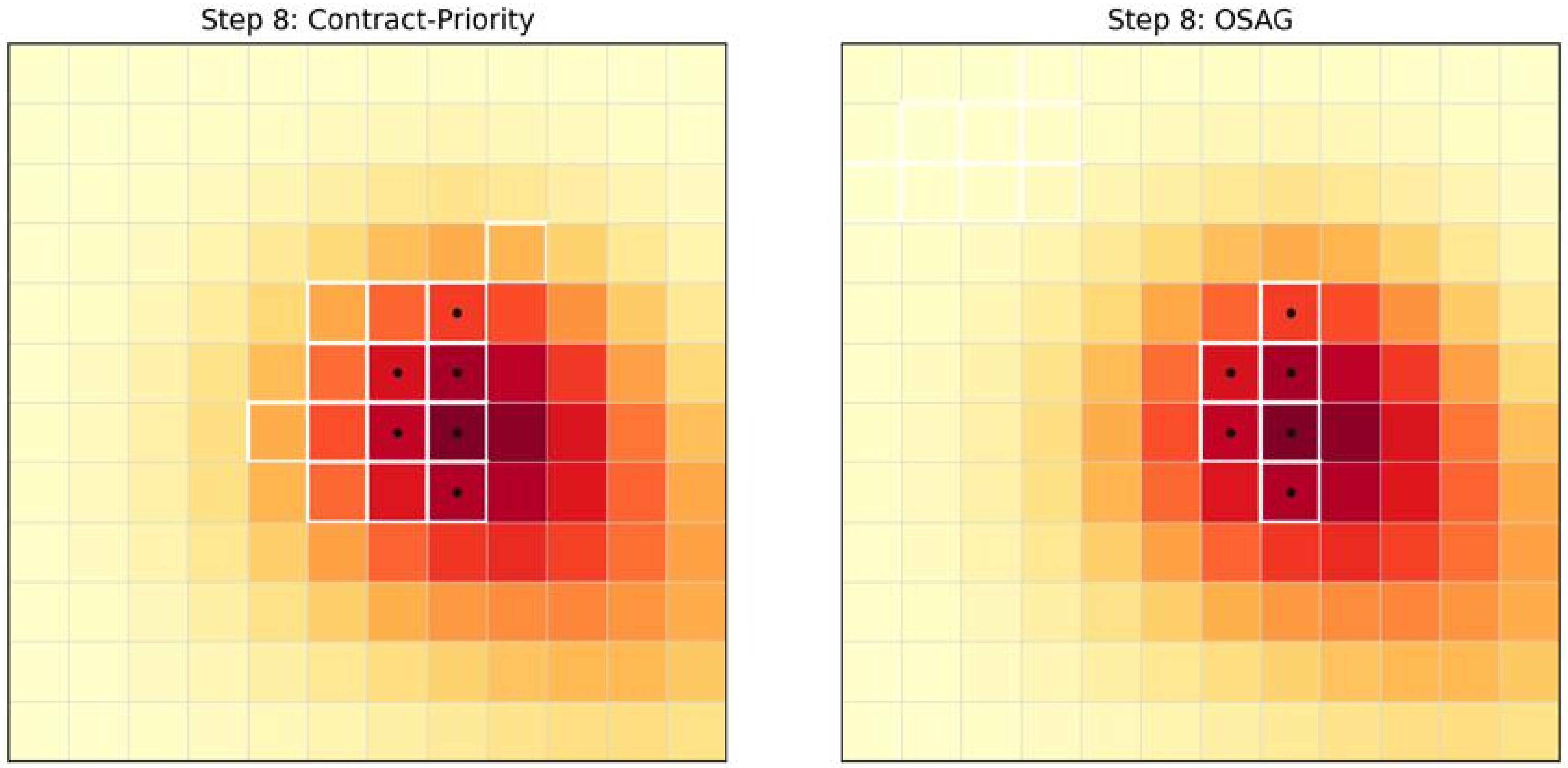
Scoped ResMLP extension

EuroSAT only. This is a robustness extension, not a second main benchmark.

Policy	Backbone	Acc_all	Acc_high	PrioCovErr
Random	Simple	86.93	73.68	23.81
Random	ResMLP	86.34	73.51	23.79
OSAG	Simple	86.39	76.05	0.20
OSAG	ResMLP	86.26	74.10	0.15

- Absolute utility shifts slightly with the stronger backbone.
- The central governance result still survives.
- This thesis is not a backbone SOTA study.

Why the Demo Matters



Why it belongs in the defense

- The benchmark chapter is the scientific evidence chain.
- The dispatch demo exposes budgets, deadlines, and missed-service consequences step by step.
- That makes the governance behavior easier to explain than a table alone.
- The demo supplements the benchmark; it does not replace it.

Operational setting

- 12 x 12 emergency grid over 10 decision steps.
- Observation budget = 14 tiles and annotation budget = 6 tiles per step.
- Six contracts with explicit priorities and deadlines.
- Policies shown: Random, Contract-Priority, and OSAG.

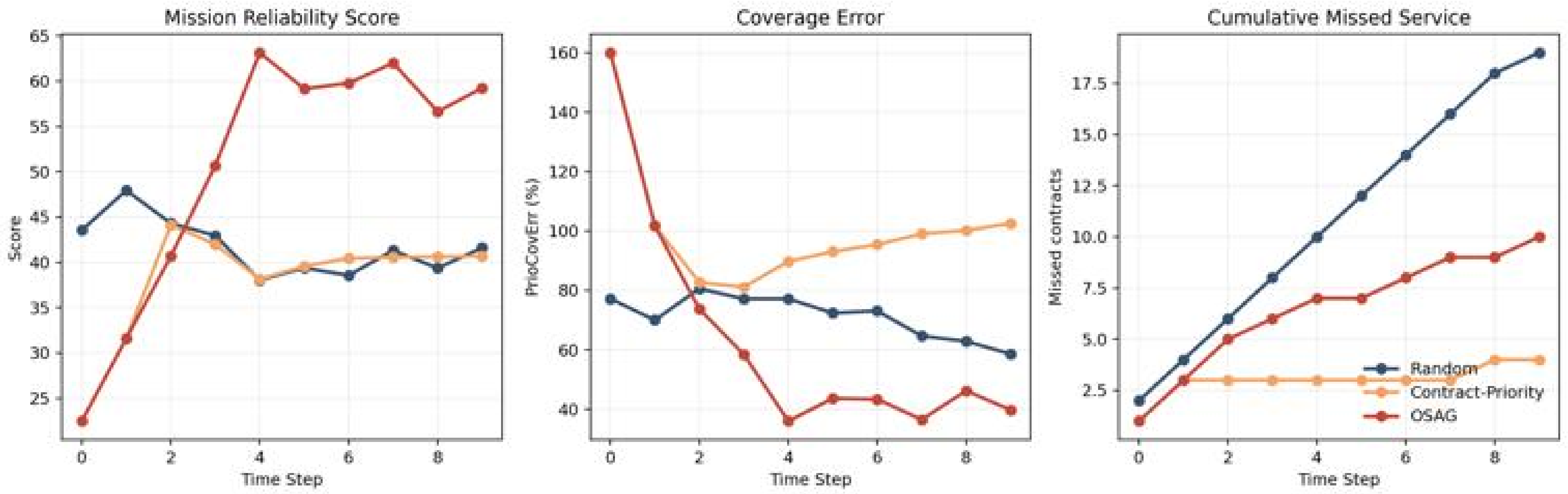
Dispatch Demo Results

End-of-run summary

Policy	Reliability	Compliance	Coverage Err	Missed
Random	44.12	53.67%	0.586	19
Contract-Priority	137.49	22.91%	1.026	4
OSAG	106.78	60.96%	0.397	10

Interpretation

- OSAG is best on the governance-oriented summary: reliability, compliance, and coverage error.
- Contract-Priority is only attractive if the user over-focuses on missed service alone.
- The sensitivity audit from saved outputs keeps OSAG top-ranked around the thesis weighting.



Repository and Code Architecture

Formal benchmark path

- Start from scripts/run_all_real_experiments.py.
- The core logic lives in scripts/reproduce_osag_real.py.
- Important functions to mention: build_contract_table_from_meta, run_one, and compute_coverage_errors.
- That path generates logs, fresh CSVs, figures, and runtime summaries.

Dispatch-demo path

- run_visual_demo.py is the simple entry for defense day.
- run_dispatch_demo.py builds the saved outputs.
- dispatch_demo_core.py contains choose_observations and the contract table.
- dispatch_demo_assets.py writes the dashboard and timeline visuals.

Safe files to open live

```
scripts/run_all_real_experiments.py
scripts/reproduce_osag_real.py
  build_contract_table_from_meta(...)
  compute_coverage_errors(...)
osag_demo/run_dispatch_demo.py
osag_demo/dispatch_demo_core.py
  choose_observations(...)
```

Defense point: the committee does not need a large raw code dump. They need a clear path from entry script to core logic and then to saved outputs.

Limitations and Claim Boundaries

What the thesis supports

- Governance-aware sampling can strongly reduce contract-level policy misalignment on the current HSI and EuroSAT MSI benchmarks.
- The effect survives fresh reruns, a contract-design ablation, runtime scrutiny, a scoped ResMLP extension, and an operational demo.
- The reproducibility appendix makes the evidence chain auditable through scripts, configs, manifests, logs, and saved outputs.

What is not being claimed

- The HSI split is pixel-stratified rather than spatial-blocked, so absolute accuracy may be optimistic relative to strict deployment conditions.
- The 'graph' is a modeling view, not a full explicit graph optimizer in the current implementation.
- The main benchmark uses a lightweight MLP to isolate governance effects; the thesis is not a backbone SOTA study.
- The ResMLP extension is scoped to EuroSAT only.
- The demo is supportive evidence, not the main benchmark.
- Provenance is documented through manifests and stage reports rather than a git commit ledger inside the thesis workspace.

Reproducibility and Public Release

Fully rerun main benchmark

HSI and EuroSAT MSI main tables are fresh five-seed reruns from formal scripts and canonical local inputs.

Fully rerun extensions

EuroSAT coarse-vs-fine ablation is a fresh three-seed rerun; the EuroSAT ResMLP extension is a fresh five-seed rerun.

Artifact-level reproducibility

Dispatch demo outputs, thesis figures/tables, and the thesis PDF can all be regenerated locally.

Evidence boundary and companion release

- Historical conference CSV files remain in the workspace only for traceability checks. They are not presented as fresh evidence.
- The thesis workspace is not a git repository, so provenance is documented through manifests, configuration snapshots, saved outputs, and stage reports.
- Companion public source release: https://github.com/dqswordman/osag-eo-governance

Final Takeaways

1. The thesis contribution is governance-aware EO training.

OSAG makes contract-level service allocation explicit instead of leaving it to raw sample frequency.

2. The main fresh reruns support strong policy-alignment gains.

On both HSI and EuroSAT MSI, OSAG-family policies drive PrioCovErr close to zero while keeping competitive utility.

3. The defense story is broader than one table.

Benchmark reruns, demo behavior, code path, and Appendix A together make the thesis auditable and explainable.

After the slide presentation, I can move to the saved demo outputs and the key code files if the committee wants to inspect them live.

Backup: Where is the 'graph' really?

- In the current thesis, the graph is a modeling view over the contract space, not a separate graph neural network or explicit graph optimizer.
- The graph intuition matters because contracts can be related through semantic similarity, spatial adjacency, or hierarchical refinement.
- Chapter 3 uses this graph view to motivate why contract design changes governance cost, especially in the EuroSAT coarse-vs-fine study.
- Future work could instantiate the graph explicitly through region adjacency graphs or learned semantic neighborhoods.

Safe oral answer

graph = structured contract space

The current implementation is mainly a contract-aware sampling and monitoring layer.

Do not overclaim

not a full graph optimizer

The thesis deliberately treats explicit graph algorithms as future work.

Backup: HSI Split and Spatial Leakage Boundary

What the pipeline does

- Corrected Indian Pines and corrected Salinas are aligned to a common spectral subset and merged into a joint HSI setting.
- Labeled pixels are flattened first, then assigned by a class-stratified 60/20/20 train/validation/test split.
- This is pixel-stratified, not spatial-blocked.

How to answer the leakage question

- Yes, neighboring pixels from the same original scene can appear in different splits.
- That means absolute accuracy may be optimistic relative to a stricter spatial-blocked evaluation.
- The main thesis claim should therefore be read as a comparative governance result under a fixed benchmark protocol.
- The benchmark still remains useful because all compared policies share the same split and the governance effect is large and consistent.

Backup: Demo Metrics, Q_high, K, and Sensitivity

Metric interpretation

- Reliability is a hand-set composite of compliance, coverage error, missed service, and critical capture K.
- Q_high is separate: it is the final mean high-priority model-quality state, not the same quantity as K.
- Coverage error in the demo is an operational contract-level metric analogous to, but not identical with, benchmark PrioCovErr.
- Reliability is useful as a dashboard summary, but it does not replace the primitive metrics.

Sensitivity audit

- No new simulation was rerun for the sensitivity note.
- The audit recomputed rankings from saved dispatch CSV outputs only.
- OSAG stays top-ranked under equal weights and several local perturbations around the thesis weighting.
- Only more extreme capture-dominated weightings favor Contract-Priority.